

COMPUTER NETWORK PRACTICUM
PING AND ROUTE



KHEN MUHAMMAD CAHYO

244107023002

TI-21

STATE POLYTECHNIC OF MALANG
INFORMATION TECHNOLOGY DEPARTMENT

PING PRACTICUM

- **10.10.10.1 PING**

```
debian@debian:~$ ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.602 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=0.729 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.687 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.691 ms
64 bytes from 10.10.10.1: icmp_seq=5 ttl=64 time=0.692 ms
64 bytes from 10.10.10.1: icmp_seq=6 ttl=64 time=0.746 ms
^C
--- 10.10.10.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5086ms
rtt min/avg/max/mdev = 0.602/0.691/0.746/0.045 ms
debian@debian:~$ _
```

- **10x PING DETIK.COM**

```
debian@debian:~$ ping -c 10 detik.com
PING detik.com (203.190.242.211) 56(84) bytes of data.
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=1 ttl=51 time=17.0 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=2 ttl=51 time=16.8 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=3 ttl=51 time=17.0 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=4 ttl=51 time=17.4 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=5 ttl=51 time=17.1 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=6 ttl=51 time=17.6 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=7 ttl=51 time=17.3 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=8 ttl=51 time=17.3 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=9 ttl=51 time=16.9 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=10 ttl=51 time=17.6 ms
--- detik.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9070ms
rtt min/avg/max/mdev = 16.791/17.208/17.646/0.264 ms
debian@debian:~$
```

- **5x PING KOMPAS.COM WITH 3s PAUSE**

```
debian@debian:~$ ping -c 5 -i 3 kompas.com
PING kompas.com (3.171.198.98) 56(84) bytes of data.
64 bytes from server-3-171-198-98.sin2.r.cloudfront.net (3.171.198.98): icmp_seq=1 ttl=241 time=27.8 ms
64 bytes from server-3-171-198-98.sin2.r.cloudfront.net (3.171.198.98): icmp_seq=2 ttl=241 time=25.7 ms
64 bytes from server-3-171-198-98.sin2.r.cloudfront.net (3.171.198.98): icmp_seq=3 ttl=241 time=26.6 ms
64 bytes from server-3-171-198-98.sin2.r.cloudfront.net (3.171.198.98): icmp_seq=4 ttl=241 time=25.8 ms
64 bytes from server-3-171-198-98.sin2.r.cloudfront.net (3.171.198.98): icmp_seq=5 ttl=241 time=25.9 ms
--- kompas.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 12063ms
rtt min/avg/max/mdev = 25.749/26.378/27.795/0.772 ms
debian@debian:~$ _
```

- **5x PING MICROTIC.COM WITH 0.3s TIMEOUT**

```
debian@debian:~$ ping -c 5 -W 0.3 microtic.com
PING microtic.com (159.148.147.252) 56(84) bytes of data.
64 bytes from 159.148.147.252 (159.148.147.252): icmp_seq=1 ttl=42 time=215 ms
64 bytes from 159.148.147.252 (159.148.147.252): icmp_seq=2 ttl=42 time=215 ms
64 bytes from 159.148.147.252 (159.148.147.252): icmp_seq=3 ttl=42 time=216 ms
64 bytes from 159.148.147.252 (159.148.147.252): icmp_seq=4 ttl=42 time=215 ms
64 bytes from 159.148.147.252 (159.148.147.252): icmp_seq=5 ttl=42 time=215 ms
--- microtic.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4015ms
rtt min/avg/max/mdev = 214.568/214.947/215.562/0.365 ms
debian@debian:~$ _
```

ROUTE PRACTICUM

- **Route List**

```
debian@debian:~$ ip route show
default via 10.10.10.1 dev ens3 onlink
10.10.10.0/24 dev ens3 proto kernel scope link src 10.10.10.24
debian@debian:~$ _
```

- **Delete Default IP Gateway**

```
debian@debian:~$ sudo ip route del default
debian@debian:~$ ip route show
10.10.10.0/24 dev ens3 proto kernel scope link src 10.10.10.24
debian@debian:~$
```

- **Test PING After Delete Default IP Gateway**

```
debian@debian:~$ ping 192.168.60.142
ping: connect: Network is unreachable
debian@debian:~$ ping detik.com
ping: connect: Network is unreachable
debian@debian:~$ _
```

The above result occurs because the computer loses the main route to access the outside network (including the internet) so the system does not know where to send packets.

- **Add Default IP Gateway**

```
debian@debian:~$ sudo ip route add default via 10.10.10.1
debian@debian:~$ ip route show
default via 10.10.10.1 dev ens3
10.10.10.0/24 dev ens3 proto kernel scope link src 10.10.10.24
debian@debian:~$ _
```

- **Test PING After Add Default IP Gateway**

```
debian@debian:~$ ping 192.168.60.142
PING 192.168.60.142 (192.168.60.142) 56(84) bytes of data.
From 192.168.60.24 icmp_seq=1 Destination Host Unreachable
From 192.168.60.24 icmp_seq=2 Destination Host Unreachable
From 192.168.60.24 icmp_seq=3 Destination Host Unreachable
From 192.168.60.24 icmp_seq=4 Destination Host Unreachable
^C
--- 192.168.60.142 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4065ms
pipe 4
debian@debian:~$ ping detik.com
PING detik.com (203.190.242.211) 56(84) bytes of data.
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=1 ttl=51 time=15.0 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=2 ttl=51 time=15.3 ms
64 bytes from s2-211-242.190.203.detik.com (203.190.242.211): icmp_seq=3 ttl=51 time=14.8 ms
^C
--- detik.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2016ms
rtt min/avg/max/mdev = 14.767/15.001/15.254/0.199 ms
debian@debian:~$
```

- **Block Access From Range IP 10.10.10.0/24**

```
debian@debian:~$ sudo iptables -A INPUT -s 10.10.10.0/24 -j DROP
debian@debian:~$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP       all  --  10.10.10.0/24          0.0.0.0/0

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
debian@debian:~$
```